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Machine for separating packagings

Abstract

A machine for separating packagings includes a transport belt which is inclined longitudinally at a first angle and laterally at a second angle so that a first longitudinal side is positioned higher than a second longitudinal side. The transport belt is further provided with a plurality of flights which are arranged near the second longitudinal side.

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Background/Summary

(1) This claims priority from Italian Application No. 102022000014002, filed Jul. 1, 2022, the disclosure of which is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

(2) The present invention relates to a machine for separating packagings.

(3) The invention has been developed with particular regard, though in a non-limiting manner, for a machine for manipulating packagings, particularly for pharmaceuticals, in order to allow one packaging to be removed at a time from a group of packagings arranged in bulk.

Technological Background and Summary

(4) There have existed for some time automatic stores for storing and managing pharmaceuticals, typically inside a pharmacy or a hospital. So that these stores can correctly manage the cataloguing and the insertion of new pharmaceuticals, they need to receive at the inlet the packagings in an

organized manner, preferably one at a time. The product supplies typically arrive in boxes, the content of which has to be separated in order to be sorted and conveyed to the store automatically. (5) There are known different machines which, from a disordered group of packagings, remove them one at a time therefrom. For example, it is known to provide a plurality of transport belts in a cascading manner, each one having a different advance speed, in an increasing manner. In this manner, during the movement between a first belt and a second belt, two packagings which are very close together are slightly spaced apart and the spacing thereof increases with each new transfer between two adjacent transport belts.

(6) A problem of the machines of the known type is that, when two packagings are very near, at the limit completely alongside each other, the known solution is not capable of separating them. In this manner, there will also be two packagings which are placed one beside the other at the outlet from the machinery. This requires the provision of an additional system for overcoming the inevitable errors of the machinery, for example, a vision system which is capable of verifying the presence of several packagings beside each other and a system for removing them from the transport belt. These solutions are naturally complex and costly.

(7) Another problem of the known machines described above is that they are not generally suitable for processing cylindrical packagings, for example, packagings of unguents, cremes, tablets, capsules, etc., because they roll on the transport belts, preventing correct spacing thereof.

(8) There also exist automatic sorting systems which use mechanical arms to grip the packagings one by one. However, they are very expensive and sometimes also slow.

(9) An object of the invention is to solve the problems of the prior art and in particular to provide a more precise and effective machine for separating packagings than the existing ones. Another object is to provide a machine which is also suitable for separating cylindrical packagings. Another object is to provide a mechanically simple device which is reliable to use and which has low installation and maintenance costs.

(10) According to a first aspect, there is described a machine for separating packagings comprising a transport belt which can be inclined longitudinally at a first angle. The longitudinal direction is defined as the direction in which the transport belt transports packagings which are placed thereon. The transport belt may comprise an inlet and an outlet and the inlet is positioned lower than the outlet. The transport belt can be inclined laterally at a second angle. The transport belt may have a first and a second longitudinal side, the first longitudinal side being positioned higher than the second longitudinal side. The transport belt can be provided with a plurality of flights which are arranged near the second longitudinal side.

(11) The transport belt with a double longitudinal and transverse inclination allows the packagings to be separated in a much more effective manner than the one carried out by a planar transport belt or a belt which in any case does not have any lateral inclination because the lateral inclination promotes an alignment of the packagings in a row, in the region of the second longitudinal side.

(12) The longitudinal inclination angle is preferably between 30° and 45° . In this manner, the inclination is sufficient to prevent packagings from being able to be pushed on the transport belt simply by friction on a support plane of the transport belt. At the same time, it is not thereby raised so as to cause rolling of the objects which are positioned on the belt.

(13) The transverse inclination angle is preferably between 5° and 15° . In this manner, the inclination is sufficient to cause the packagings to slide laterally and slowly in order to be able then to be taken by the flights.

(14) According to an aspect, there is described a machine for separating packagings comprising a transport belt which is provided with flights which have a small transverse dimension with respect to a transverse dimension of the transport belt. Preferably, the flights have a small transverse dimension, at least less than $\frac{1}{3}$ of the transverse dimension of the transport belt. In this manner, the transport of two juxtaposed packagings is prevented. Furthermore, the transverse dimension of the flight is equal at least to the radius of the greatest cylindrical container which it is desirable to

transport. Preferably, each flight has a height, when measured in a perpendicular direction with respect to a support plane which is defined by the transport belt, which is slightly less than the minimum dimension of the box which it is desirable to move.

(15) Preferably, at least one flight has a height greater than the transverse dimension of the flight itself and proportional to the radius of the greatest cylinder which it is desirable to move.

(16) According to another aspect, there is described a machine for separating packagings comprising a transport belt which is provided, in the region of an inlet thereof, with a base wall which is movable with cyclical translational and/or rotational movement.

(17) According to another aspect, there is described a machine for separating packagings comprising a transport belt, wherein the belt may be monolithic. The flights are therefore formed in one piece with the belt itself.

(18) According to another aspect, there is described a machine for separating packagings comprising at least two transport belts which are positioned in a cascading manner with respect to each other.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Additional features and advantages will be appreciated from the following detailed description of a preferred embodiment of the invention with reference to the appended drawings which are provided purely by way of non-limiting example and in which:

(2) FIG. 1 is a perspective view of a machine for separating packagings,

(3) FIG. 2 is a side view of the machine of FIG. 1,

(4) FIG. 3 is a rear view of the machine of FIG. 1,

(5) FIG. 4 is a view, drawn to an enlarged scale, of a detail of the lower portion of a first transport belt of the machine of FIG. 1, and

(6) FIG. 5 is a view, drawn to an enlarged scale, of a detail of the upper portion of the first transport belt of the machine of FIG. 1.

DETAILED DESCRIPTION

(7) Now with reference to the drawings, a machine **1** for separating packagings comprises a carrier structure **10**, which is schematically depicted in FIG. 1. Transport belts **20**, **30** and **60** are fixed to the carrier structure.

(8) A first transport belt **20** is provided for depositing there the packagings C to be separated in bulk. The transport belt **20** is preferably horizontal or slightly inclined longitudinally at an angle α in such a manner that an inlet zone **23** and an outlet zone **25** thereof which are intended to receive packagings C from an operator and to convey packagings to the second transport belt **30**, respectively, are located one lower than the other. The angle α is preferably less than 10° .

(9) The first transport belt **20** in the embodiment depicted is provided with a plurality of flights **22**, that is to say, gripping elements which project with respect to a support plane **24** which is defined by the transport belt itself. The flights preferably have a transverse dimension which is equal to the width of the transport belt and serve to push forwards the packagings which are located on the transport belt.

(10) In another embodiment Applicant, the first transport belt **20** does not have flights. In this case, the belt preferably has a high friction coefficient and/or has surface irregularities which promote the advance of the boxes.

(11) A pair of lateral edges **26** are provided at the sides of the first transport belt **20** in order to prevent packagings from accidentally falling off the sides of the transport belt itself. A motor **28** actuates the transport belt.

(12) The second transport belt **30** comprises an inlet zone **32** and an outlet zone **34** which are

intended to received packagings C from the first transport belt **20** and to convey packagings C to the third transport belt **60**, respectively. The second transport belt **30** is actuated by a second motor **35**.

(13) As can better be seen in FIGS. **2** and **3**, in which for the sake of simplicity neither the carrier structure **10** nor the motors which move the transport belts are depicted, the second transport belt is arranged so that the inlet zone **32** is lower than the outlet zone **34**. In other words, it is inclined longitudinally at an angle α' . Furthermore, the transport belt is also inclined laterally at a second angle β' ; in this manner, a first longitudinal side **36** is positioned higher than a second longitudinal side **38**. The longitudinal inclination angle α' is preferably between 30° and 45° . The transverse inclination angle β' is preferably between 5° and 15° .

(14) In order to prevent the risk of the packagings falling, there is preferably provided at least one lateral edge **40** in the region of the longitudinal side **38**. Preferably, an additional lateral edge **42** is also provided in the region of the longitudinal side **36**.

(15) The second transport belt is provided with a plurality of flights **44**, **46** which project with respect to a support plane **48** which is defined by the transport belt **30** itself. The flights **44**, **46** have a transverse dimension which is far less than the transverse dimension of the transport belt **30**. In particular, they preferably have a width which is a maximum of $\frac{1}{3}$ of the width of the transport belt. The transverse dimension of the flight is preferably less than the width of the smallest packaging which it is intended to move in order to reduce to the maximum extent the risk of transporting two packagings alongside each other. The transverse dimension of the flight is at least equal to the radius of the greatest cylindrical container which it is desirable to transport in order to allow the largest packagings also to be moved in an optimum manner.

(16) The height of the flights, which is measured in a perpendicular direction with respect to a support plane which is defined by the transport belt, is slightly less than the minimum dimension of the box which it is desirable to move.

(17) During use, the flights raise the packagings which are positioned in the inlet zone **32** of the second transport belt **30**, pushing them towards the outlet **34** and therefore towards the third transport belt **60**.

(18) The flights are preferably of two types. A first type of flight **44**, which is preferably predominant in terms of numbers, has a height, which is measured in a direction perpendicular to the support plane **48**, less than or in any case approximately equal to the width thereof. The flights **44** of the first type are preferably in the shape of a parallelepiped. These flights **44** are arranged near the second longitudinal side **38**, that is to say, the lowest longitudinal side, towards which the packagings which are positioned on the transport belt slide, during use, as a result of the effect of the inclination at an angle β .

(19) A second type of flights **46** according to the depicted embodiment, which can clearly be seen in detail in FIGS. **4** and **5**, has a more complex form: there is fixed to a parallelepiped, preferably with a form and dimensions identical to the other flights **44** provided on the belt, a projection which is approximately in the form of a prism with a triangular base and which is orientated so that the two triangular faces are arranged perpendicularly to the support plane **48** of the transport belt **30**. Preferably, the triangular faces are in the form of a rectangular or right triangle and are arranged in such a manner that one cathetus is located in the support plane **48** and the other one is directed towards the outlet **34**. The flight **46** is anchored to the mat by means of the parallelepiped mentioned above, therefore it is fixedly joined to the web only by a surface which is indicatively equal to that of the flights **44**.

(20) More generally, the flights of the second type **46** have a height, which is measured in a direction perpendicular to the support plane **48**, which is far greater than the width thereof. Furthermore, they have an edge **50** which is preferably perpendicular to the support plane and which is able to strike, during use, against packagings which are arranged in the inlet zone **32** of the transport belt **30**.

(21) Preferably, the web of the transport belt **30** is monolithic, that is to say, the flights are fused to the web itself, not bonded, welded or otherwise fixed. In this manner, the risk of detachment is prevented, with the machine being blocked as a result and repairs being necessary.

(22) In the region of the lower end of the transport belt a movable base wall **52** is provided. In the embodiment depicted, the movable base wall **52** is moved by means of a cam **54**. It can be configured so as to move in translation in the advance direction of the transport belt, in both directions. Alternatively, it can be configured to rotate about a horizontal axis, which is parallel with the plane which is defined by the same movable base wall **52**, or vertical axis. Furthermore, a rotational translational movement of the movable base wall is not excluded; more generally, the movable base wall **52** is configured in such a manner that a lower portion **56** thereof moves alternately and cyclically in the advance direction of the transport belt **30** and in the opposite direction. The movable base wall **52** has apertures **58** in order to allow the passage of the flights **44**, **46**.

(23) The third transport belt **60** is positioned in such a manner that the packagings being discharged from the second transport belt **20** are conveyed by means of a chute **61** towards the third transport belt **60**.

(24) The third transport belt **60** is generally similar to the second transport belt described above. It comprises an inlet zone **62** and an outlet zone **64** which are intended to receive packagings from the second transport belt **30** and to convey packagings being discharged from the machine, respectively, for the subsequent automated storage operations of the packagings. The third transport belt **60** is actuated by a third motor **65**.

(25) The third transport belt is also inclined longitudinally at an angle α'' . Furthermore, the transport belt is also inclined laterally at a second angle β'' . The longitudinal inclination angle α'' is preferably between 30° and $^\circ$. The transverse inclination angle β'' is preferably between 5° and 15° . The angles α'' and β'' can be identical to the angles α' and β' of the second transport belt **30**. As a result of the effect of the lateral inclination, a first longitudinal side **66** is positioned higher than a second longitudinal side **68**. Lateral edges **70**, **72** are preferably provided at the longitudinal sides **68**, **66**.

(26) As described above with reference to the second transport belt, the third transport belt is also provided with a plurality of flights **44**, **46** which project with respect to a support plane **78** which is defined by the transport belt **60**. The flights **44**, **46** of the third transport belt are generally similar to the flights **44**, **46** of the second transport belt, to the description of which reference may be made.

(27) In the region of the lower end of the transport belt there is provided a movable base wall **52** which is similar to the one described above for the second transport belt **30**.

(28) The machine further comprises an asynchronous control system for the transport belts which allows optimization of the charge of packagings on the belts. According to the embodiment depicted, which is provided merely by way of non-limiting example, this control system comprises a plurality of presence detectors, for example, photo-cells, in order to detect the presence of packagings in predetermined positions.

(29) A first pair of presence detectors (FIG. 4) comprises a detector **100** which is positioned in the outlet zone **25** of the first transport belt **20** and a detector **102** which is positioned at the inlet **32** of the second transport belt **30**.

(30) During use, if the detector **102** does not detect any presence at the inlet **32** of the second transport belt **30**, the first transport belt is actuated by actuating the motor **28** in order to bring a new packaging onto the second transport belt.

(31) If, instead, the detector **102** already detects a presence at the inlet **32** of the second transport belt **30**, the first transport belt is actuated only if the detector **101** does not detect any presence at the outlet **24** of the first transport belt **20** in order to bring a new packaging to the outlet **24**, ready to be conveyed to the second transport belt as soon as the detector **102** signals the absence of packagings at the inlet **32** of the second transport belt **30**.

(32) Similarly, an additional pair of detectors **104, 106** which can be seen in FIG. 5 are provided in the outlet zone **34** of the second transport belt **30** and at the inlet **62** of the third transport belt **60**, respectively, with operation which is generally similar to that of the detectors **100, 102**.

(33) An additional pair of detectors may also be provided for managing the passage from the third transport belt **60** to a subsequent processing station, which does not form part of the present machine. In this case, a detector may be provided at the outlet from the third transport belt **60** in order to collaborate with a detector which is provided at the inlet to the subsequent processing station.

(34) During use, a plurality of packagings **C** are deposited, manually or by another machine, on the first transport belt **20**. The first transport belt, on the basis of the signals which are detected by the presence detectors **100, 102**, is actuated until allowing at least one packaging to fall on the second transport belt **30**. If a plurality of packagings are gathered together, as in the example of FIG. 4, often more packagings are allowed to fall together, forming small piles **P** of packagings on the second transport belt **30**.

(35) The movement of the base wall **52**, together with the lateral inclination of the transport belt **30**, cause the packagings **C** to slide towards the lateral edge **40**. Once in contact with the lateral edge **40**, the flights **44** take the packagings.

(36) The flights **46** of the second type are particularly useful on different occasions. Particularly, as can be seen in FIG. 4, they allow cylindrical packagings **C'** which are arranged with the horizontal axis thereof to be rotated. A horizontal cylindrical packaging could not in fact be collected by a flight **44** or by another conventional flight: because a flat surface which can rest on the flight **44** is not present, during the passage of the flights **44** it passes them but without being urged towards the outlet and actually blocking the machine. When, however, a cylindrical packaging which is arranged with the horizontal axis thereof is struck by a flight **46**, the flight rotates it and arranges it so as to have the axis parallel with the longitudinal direction of the transport belt, allowing the subsequent flight **44** to collect it. FIG. 4 shows a cylindrical packaging **C'** while it is being rotated and another packaging **C'** is being transported by a flight **44**.

(37) In the event that there are stacks **P** formed by numerous superimposed packages at the entrance **32**, with the formation of bridges as well, the flights of the second type **46** allow the packagings which are piled to fall and then proceed to separation.

(38) Then, as a result of the minimal transverse dimensions of the flights **44, 46**, it is possible to raise a minimum number of packagings at one time. At most, it may happen to lift a few packagings on top of each other, but hardly two packagings side by side. Two superimposed packagings are subsequently readily separated during the passage from the second transport belt to the third (FIG. 5) because the one positioned above is conveyed to the slide **61** before the one below.

(39) Furthermore, the second and third transport belts are not aligned with each other, but instead parallel and connected by an angled chute. It is precisely the lack of alignment between the second and third transport belts and the particular design of chute **61** that mean that even if two superimposed packagings on the second transport belt were conveyed at the same time to the third transport belt, they would arrive there side by side. Because the third transport belt also has the flights **42** with small transverse dimensions, the separation of a plurality of juxtaposed packagings is immediate and effective.

(40) Naturally, the principle of the invention remaining the same, the forms of embodiment and details of construction may be varied widely with respect to those described and illustrated, without thereby departing from the scope of the invention.

(41) For example, the description is given with particular reference to packagings of pharmaceuticals, but naturally the machine is also suitable for separating other packagings or, more generally, rigid products.

Claims

1. A machine for separating packagings, the machine comprising a transport belt inclined longitudinally at a first angle and laterally at a second angle so that a first longitudinal side is positioned higher than a second longitudinal side, the machine further comprising a plurality of flights arranged adjacent the second longitudinal side of the transport belt and the flights have a transverse dimension less than a transverse dimension of the transport belt.
 2. The machine according to claim 1, wherein the first angle is between 30° and 45°.
 3. The machine according to claim 1, wherein the second angle is between 5° and 15°.
 4. The machine according to claim 1, wherein the transport belt is monolithic.
 5. The machine according to claim 1, further comprising an additional transport belt positioned in a cascading manner with respect to the transport belt.
 6. A machine for separating packagings, the machine comprising a transport belt inclined longitudinally at a first angle and laterally at a second angle so that a first longitudinal side is positioned higher than a second longitudinal side, the machine further comprising a plurality of flights arranged adjacent the second longitudinal side of the transport belt, and at least one flight of the plurality of flights has a height, when measured in a perpendicular direction with respect to a support plane defined by the transport belt, greater than a transverse dimension of the at least one flight.
 7. The machine according to claim 6, wherein the first angle is between 30° and 45°.
 8. The machine according to claim 6, wherein the second angle is between 5° and 15°.
 9. The machine according to claim 6, wherein the transport belt is monolithic.
 10. The machine according to claim 6, further comprising an additional transport belt positioned in a cascading manner with respect to the transport belt.
 11. A machine for separating packagings, the machine comprising a transport belt inclined longitudinally at a first angle and laterally at a second angle so that a first longitudinal side is positioned higher than a second longitudinal side, the machine further comprising a plurality of flights arranged adjacent the second longitudinal side of the transport belt, and a base wall movable with cyclical translational and/or rotational movement, the base wall being disposed at an inlet of the transport belt.
 12. The machine according to claim 11, wherein the first angle is between 30° and 45°.
 13. The machine according to claim 11, wherein the second angle is between 5° and 15°.
 14. The machine according to claim 11, wherein the transport belt is monolithic.
 15. The machine according to claim 11, further comprising an additional transport belt positioned in a cascading manner with respect to the transport belt.
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